

Classical Ciphers

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Introduction to the Hill Cipher

Definition: The Hill Cipher is a polygraphic substitution cipher based on linear algebra. It encrypts blocks of text by applying a matrix as a key.

Inventor: Developed by mathematician Lester S. Hill in 1929.

Key Concept: Uses matrix multiplication for encryption and decryption, making it unique among classical ciphers.

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Introduction to the Hill Cipher

Matrix: A rectangular array of numbers arranged in rows and columns.

Modular Arithmetic: Performing calculations with numbers "wrapped around" a fixed value (modulus). Hill Cipher typically uses modulo 26 for English letters.

Invertible Matrix: A matrix that has an inverse, which is essential for decryption in the Hill Cipher.

Hill Cipher Encryption Process

Select a Key Matrix:

- Choose a square matrix (e.g., 2x2 or 3x3) as the key for encryption.
- The matrix must be invertible modulo 26 to be used for decryption.

Convert Plaintext to Numerical Values:

- Assign each letter of the alphabet a number (A=0, B=1, ..., Z=25).
- Divide the plaintext into blocks of the same size as the key matrix.

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

Hill Cipher Encryption Process

Multiply the Key Matrix by the Plaintext Vector:

- For each block of plaintext, represent it as a column vector and multiply it by the key matrix.
- Apply mod 26 to keep results within the 0-25 range.

Convert the Resulting Numbers Back to Letters:

- The resulting matrix from multiplication is converted back into letters to form the ciphertext.

Example: Encrypt the plaintext "HI" using a 2x2 key matrix

1. **Plaintext**: "HI"
2. **Numeric Representation**:
    1. H = 7, I = 8
    2. Plaintext Vector = [7,8]
3. **Key Matrix**:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Hill Cipher Encryption Process

4. **Matrix Multiplication**:

$$C = K \times P \pmod{26}$$

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} (3 \times 7 + 3 \times 8) \mod 26 \\ (2 \times 7 + 5 \times 8) \mod 26 \end{bmatrix}$$

- First row: $(3 \times 7 + 3 \times 8) \mod 26 = (21 + 24) \mod 26 = 45 \mod 26 = 19$
- Second row: $(2 \times 7 + 5 \times 8) \mod 26 = (14 + 40) \mod 26 = 54 \mod 26 = 2$
- **Ciphertext Vector**: $[19, 2]$

5. **Convert Back to Letters**:
   - 19 = T, 2 = C
   - **Ciphertext**: "TC"

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HILL CIPHER (Encryption)

2 X 2 MATRIX

P = HELP (**HE LP**) $K=\begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$ C = K * P MODE 26

$P = HE = \begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$

$K * P = \begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}\begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$ MOD 26

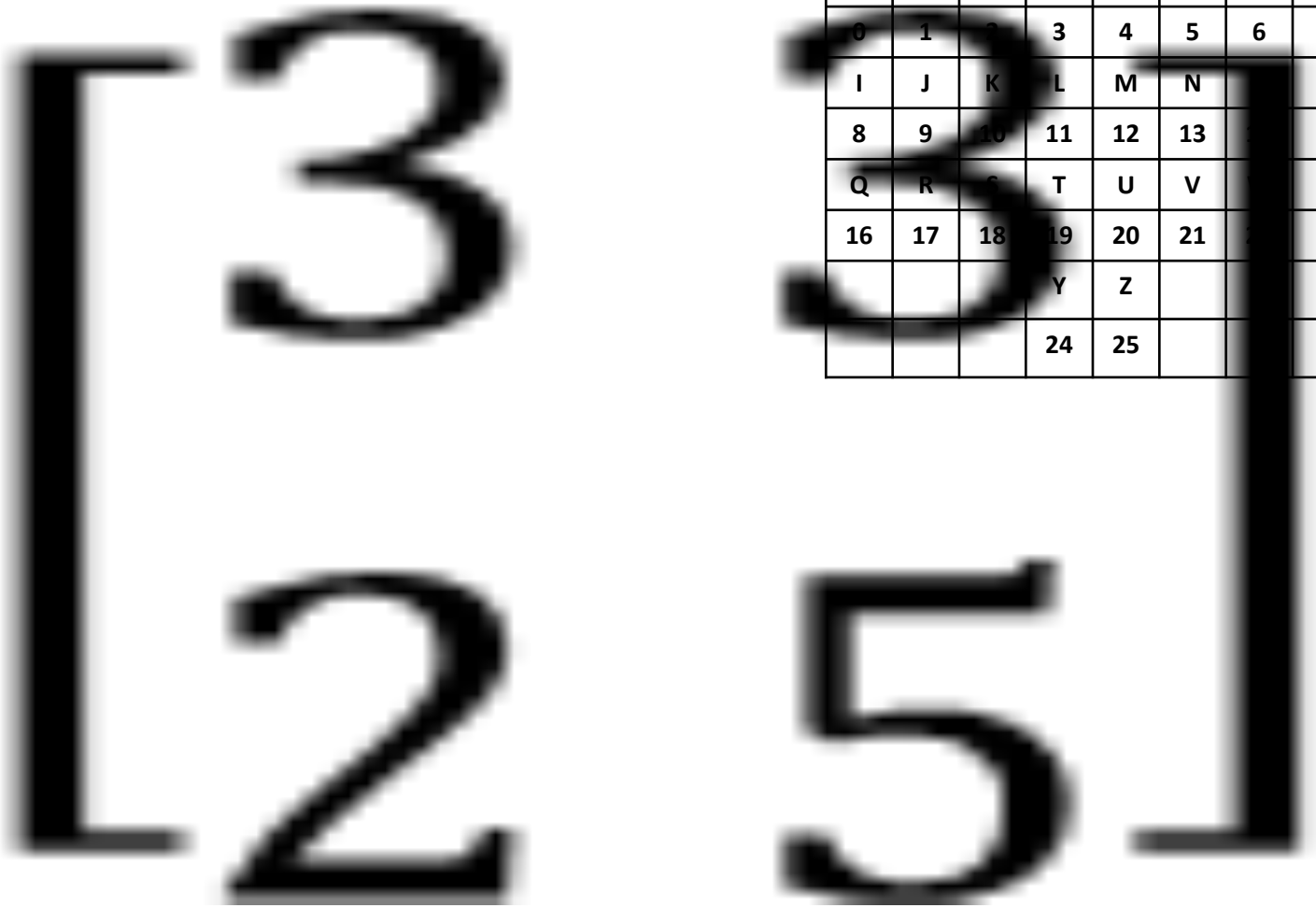
$= \begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$ MOD 26 $= \begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$

$= \begin{bmatrix}3 & 3\\ 2 & 5\end{bmatrix}$

A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

P= HELP C = HIAT

SUBSTITUTION TECHNIQUES



A	B	C	D	E	F	G	H
0	1	2	3	4	5	6	7
I	J	K	L	M	N	O	P
8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X
16	17	18	19	20	21	22	23
			Y	Z			
			24	25			

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HILL CIPHER (Encryption) Another Example

- We have to encrypt the message ‘ACT’ (n=3). The key is ‘GYBNQKURP’, which in the form of an nxn matrix looks like below:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

- ‘ACT’ is written in the form of the following vector:

$$\begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$

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- The resulting enciphered vector will be:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix} \equiv \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \pmod{26}$$

- This results in the ciphertext of 'POH'.

Hill Cipher Decryption Process

Calculate the Inverse of the Key Matrix:

- Use the key matrix and find its inverse modulo 26.The inverse matrix will reverse the encryption process.

Multiply the Ciphertext Vector by the Inverse Key Matrix:

- Convert the ciphertext into a vector.
- Multiply by the inverse key matrix, then apply mod 26.

Convert Back to Letters:

- The resulting vector from multiplication is converted back to letters to reveal the plaintext.

Hill Cipher Decryption Process

Example: Decrypt the ciphertext "TC" with the inverse of the key matrix

1. **Ciphertext**: "TC"
2. **Numeric Representation**:
    1. T = 19, C = 2
    2. Ciphertext Vector = [19,2]

3. **Inverse Key Matrix**:

$$\text{Given } K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

Hill Cipher Decryption Process

4. **Matrix Multiplication**:

$$P = K^{-1} \times C \,(\text{mod } 26)$$

$$P = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \times \begin{bmatrix} 19 \\ 2 \end{bmatrix} = \begin{bmatrix} (15 \times 19 + 17 \times 2) \mod 26 \\ (20 \times 19 + 9 \times 2) \mod 26 \end{bmatrix}$$

- First row: $(15 \times 19 + 17 \times 2) \mod 26 = (285 + 34) \mod 26 = 319 \mod 26 = 7$
- Second row: $(20 \times 19 + 9 \times 2) \mod 26 = (380 + 18) \mod 26 = 398 \mod 26 = 8$
- **Plaintext Vector**: $[7, 8]$

5. **Convert Back to Letters**:

- 7 = H, 8 = I
- **Decrypted Plaintext**: "HI"

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HILL CIPHER (Decryption) Another Example

- For the purpose of decryption, the ciphertext will have to be turned back into a vector. Simply, multiply it by the inverse matrix of the key matrix (IFKVIVVMI in letters). The inverse of the matrix will be:
- https://www.youtube.com/watch?v=aiBgjz5xbyg
- https://youtu.be/lauh7KjEfXk?si=L8K9v3PzawEyCfgT

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}^{-1} \equiv \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \pmod{26}$$

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- For the previous Ciphertext ‘POH’, multiplying it with the inverse matrix gives:

$$\begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \equiv \begin{bmatrix} 260 \\ 574 \\ 539 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} (\text{mod } 26)$$

- The output vector gives back ‘ACT’.

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- Encryption and decryption both play a crucial role in understanding the use of Hill Ciphers in general. It is important to understand that any possible matrix in the system does not represent a key matrix. Cipher decryption instead requires an inverse key matrix.

- The determinant method can determine whether the inverse exists or not. If the value of the determinant is 0 or shares a factor other than 1, it means that the matrix does not have an inverse. As a result, one will have to find or choose a different key matrix for decryption.

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The Hill cipher offers several advantages when it comes to encrypting messages:

- **Better Security**: Unlike traditional ciphers that encrypt one letter at a time, the Hill cipher works with blocks of letters. This makes it harder for attackers to break the code using frequency analysis (a technique that analyzes how often certain letters appear).
- **Versatility**: The Hill cipher can encrypt messages that include both uppercase and lowercase letters. This makes it useful for securing a wide variety of text-based information.

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- **Increased Complexity**: The Hill cipher uses matrix math for encryption, which adds complexity and makes it more challenging for someone to decode the message without the correct key. This extra complexity boosts security.
- **Strong Keys:** The strength of the Hill cipher depends on the size and randomness of the matrix used as the encryption key. A larger, more random matrix makes the encryption stronger, making it much harder for unauthorized people to decipher the message.
- **Educational Value:** The Hill cipher is based on linear algebra, which provides a good introduction to more advanced encryption techniques. It also helps in understanding the connection between matrices and encryption.

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Security Aspects of Hill Cipher

- The Hill cipher is easier to break when using small matrices, like 2×2 matrices. In these cases, the encryption can be solved fairly easily.
- However, when dealing with larger, more complex cryptographic systems, especially those with 256 possible combinations, the Hill cipher doesn't hold up well. It's considered weak in modern cryptography.
- One of the main weaknesses of the Hill cipher is its vulnerability to known-plaintext attacks. This means that if someone has access to both the original text (plaintext) and the encrypted text (ciphertext), they can use the relationship between them to figure out the encryption key. This happens because the Hill cipher relies on simple algebra, making it easier to crack when enough information is known.

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Security Aspects of Hill Cipher

- Using larger or more complex matrix multiplications in the Hill cipher doesn't necessarily make the system more secure. While it can help with mixing or spreading out the data (called diffusion), it doesn't fully protect against attacks unless combined with more advanced techniques, like non-linear operations.
- Modern encryption methods, such as AES (Advanced Encryption Standard), use different ways to mix data to strengthen security beyond what the Hill cipher offers.
- Lester S. Hill designed a special machine for encrypting with a 6×6 matrix, which provided better security at the time. However, the machine had a major drawback: its key settings couldn't be changed, making it less practical for widespread use.

Thank You

Question and Answers

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